

WAKKANAI, Japan

WMO: 474010

Lat: 45.4153N

Lon: 141.6786E

Elev: 38

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	14.4	1.9	6.1	15.2	4.2	6.8	17.1	36.3	29.6	32.1	29.5	9.5	270	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	76.6	70.8	74.7	69.5	73.0	68.3	72.4	75.4	70.8	73.7	69.2	72.2	11.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	115.1	74.7	69.6	109.1	73.1	68.0	103.1	71.6	36.2	75.5	34.8	73.8	33.5	72.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.8	24.1	21.0	DB	9.5	80.4	3.1	2.8	7.3	82.5	5.5	84.1	3.8	85.7	1.5	87.8
			WB	8.1	73.4	3.0	2.1	6.0	74.9	4.2	76.1	2.6	77.3	0.4	78.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	23.7	24.0	30.6	40.1	48.7	55.6	63.3	67.7	63.2	52.2	38.7	28.3	(6)
(7)		DBStd	16.26	5.19	5.88	5.59	5.36	6.21	5.36	5.20	4.51	4.66	5.74	7.31	5.65	(7)
(8)		HDD50	3614	815	727	601	299	101	11	0	0	0	43	343	674	(8)
(9)		HDD65	7561	1280	1147	1066	746	507	283	95	25	88	397	788	1139	(9)
(10)		CDD50	1715	0	0	0	3	59	181	413	548	395	111	5	0	(10)
(11)		CDD65	187	0	0	0	0	0	2	43	108	34	0	0	0	(11)
(12)		CDH74	243	0	0	0	0	0	1	35	173	34	0	0	0	(12)
(13)		CDH80	7	0	0	0	0	0	0	2	4	1	0	0	0	(13)
(14)	Wind	WSAvg	10.2	11.2	10.6	10.5	10.6	10.5	9.5	8.6	8.8	9.2	10.3	11.2	11.3	(14)
(15)	Precipitation	PrecAvg	44.4	3.7	2.6	2.2	2.1	2.8	2.6	3.9	4.4	5.3	5.1	5.0	4.7	(15)
(16)		PrecMax	69.1	6.7	5.8	5.5	4.3	7.9	6.4	9.2	10.9	13.0	11.1	9.6	7.6	(16)
(17)		PrecMin	30.7	1.4	0.8	0.4	0.5	0.7	0.1	0.2	0.4	1.2	1.9	1.6	1.4	(17)
(18)		PrecStd	7.9	1.2	1.1	1.0	1.0	1.4	1.5	2.1	2.3	2.7	2.0	1.7	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	41.2	46.1	59.2	68.7	72.1	77.3	80.0	77.4	67.4	58.4	46.5	(19)
(20)			MCWB	34.6	38.2	41.5	49.7	56.2	62.3	70.3	72.9	71.2	60.9	53.5	43.0	(20)
(21)		2%	DB	35.1	38.1	42.8	54.1	63.5	68.2	74.5	77.7	74.5	65.1	55.6	42.4	(21)
(22)			MCWB	32.0	34.6	38.8	47.0	54.4	61.1	68.4	71.7	68.5	59.2	51.2	38.8	(22)
(23)		5%	DB	33.1	35.6	40.6	50.9	60.4	65.6	72.3	75.9	72.3	62.8	52.5	39.2	(23)
(24)			MCWB	30.4	32.0	37.1	45.0	53.1	60.2	67.7	70.8	66.8	57.1	47.6	35.6	(24)
(25)		10%	DB	31.2	32.8	38.7	48.0	57.6	63.3	70.6	74.3	70.3	60.7	49.9	36.3	(25)
(26)			MCWB	28.6	29.7	35.4	43.2	51.7	58.8	67.0	69.8	64.7	54.9	45.0	32.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.1	38.1	42.2	51.2	58.7	64.9	72.3	74.8	72.5	62.9	54.7	43.8	(27)
(28)			MCD B	37.7	40.6	45.3	56.6	64.1	69.4	75.7	78.1	76.0	66.5	57.7	46.0	(28)
(29)		2%	WB	32.7	35.3	39.4	47.9	56.1	62.7	70.3	73.3	70.0	60.0	51.4	39.2	(29)
(30)			MCD B	34.6	37.8	42.1	52.9	61.6	66.0	72.6	76.2	73.5	64.0	55.2	42.0	(30)
(31)		5%	WB	30.8	32.4	37.7	45.7	54.1	61.1	68.8	71.9	67.6	58.0	48.3	36.0	(31)
(32)			MCD B	32.8	35.0	40.3	49.9	58.9	64.2	71.3	74.9	71.3	62.0	51.8	38.8	(32)
(33)		10%	WB	28.7	30.0	35.7	43.7	52.4	59.5	67.4	70.5	65.6	56.1	45.4	33.0	(33)
(34)			MCD B	31.1	32.6	38.5	47.4	56.7	62.9	70.1	73.6	69.5	60.1	49.3	35.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.3	7.4	7.5	8.9	9.8	9.0	8.2	7.9	9.0	9.0	8.0	6.7	(35)
(36)			MCDBR	8.6	9.9	9.9	13.4	13.8	12.0	10.0	9.5	10.1	10.6	11.5	10.7	(36)
(37)			MCWBR	8.3	9.1	8.3	8.2	7.4	6.7	5.9	5.6	7.1	8.9	11.0	10.5	(37)
(38)		5% WB	MCD BR	8.3	9.6	9.8	12.6	12.1	10.5	8.7	8.3	9.3	10.3	12.0	10.6	(38)
(39)			MCWBR	8.4	9.4	8.8	8.4	7.5	6.7	5.8	5.4	7.5	9.5	11.9	10.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.389	0.443	0.510	0.528	0.509	0.480	0.451	0.404	0.412	0.391	0.364	(40)	
(41)		taud	2.095	1.955	1.889	1.825	1.879	2.024	2.178	2.278	2.343	2.150	2.097	2.034	(41)	
(42)		Ebn at Noon	228	238	243	239	239	245	251	254	257	230	207	201	(42)	
(43)		Edh at Noon	31	44	54	62	61	53	45	40	34	36	31	30	(43)	
(44)	All-Sky Solar Radiation	RadAvg	348	629	1070	1442	1543	1477	1414	1377	1233	823	425	270	(44)	
(45)		RadStd	31	67	89	131	171	164	179	118	115	55	47	23	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page